Lecture 10: The t Distribution

STAT 132

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1 The t Distribution

The t-distribution, often referred to as Student's t-distribution, plays a central role in statistical inference, especially when working with small sample sizes and unknown variances. Here, we will introduce the t-distribution, discuss its properties, and illustrate its connection to random samples from the normal distribution.

1.1 Definition and Properties

Definition 1.1. Let $Z \sim N(0,1)$ and $Y \sim \chi^2_m$, where Z and Y are independent. The random variable

$$T = \frac{Z}{\sqrt{Y/m}} \tag{1}$$

follows a t-distribution with m degrees of freedom (denoted as $T \sim t_m$).

Key Properties

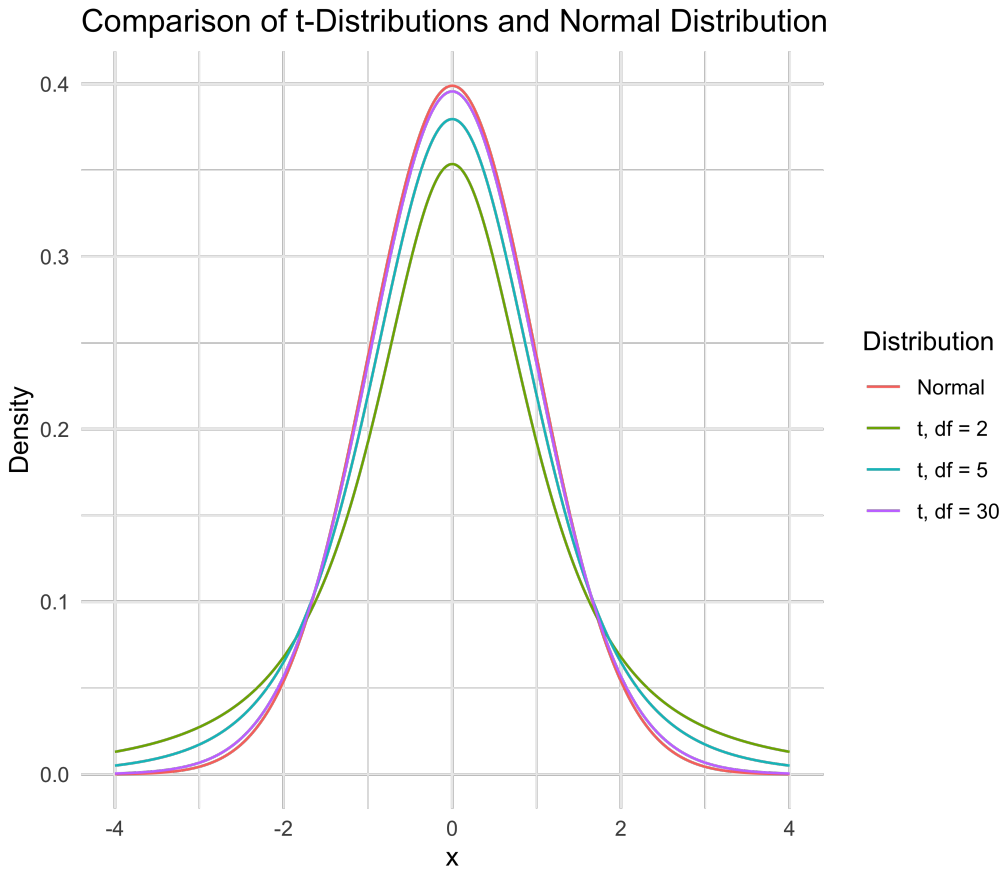
- Symmetry: This distribution is symmetric about 0, similar to the standard normal distribution.
- Moments: If $m > k$ then $\mathbb{E}(|X|^k) < \infty$. If $m \leq k$ then $\mathbb{E}(|X|^k) = \infty$.
  - For $m = 1$, the mean does not exist.
  - For $m > 1$, the mean is zero.
- Variance: For $m > 2$, $\mathrm{Var}(X) = \frac{m}{m-2}$.
- Heavier tails than the normal distribution, which decrease as degrees of freedom m increase.
- As $m \to \infty$, the t-distribution converges to the standard normal distribution.

1.2 Probability Density Function

Theorem 1.1. *The PDF of the t distribution with m degrees of freedom is*

$$f(x|m) = \frac{\Gamma\left(\frac{m+1}{2}\right)}{\sqrt{m\pi}\,\Gamma(m/2)} \left(1 + \frac{x^2}{m}\right)^{-(m+1)/2}, \quad -\infty < x < \infty.$$

1.3 Visual Comparison



2 Relationship to Normal Sampling

2.1 Key Idea

Theorem 2.1. *Suppose that $X_1, \ldots, X_n$ form a random sample from a normal distribution with mean μ and variance σ^2. Let $\bar{X}$ denote the sample mean and define*

$$S^2 = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \bar{X})^2.$$

Then, the quantity

$$U = \frac{\sqrt{n}(\bar{X} - \mu)}{\sqrt{S^2}}$$

will follow the t distribution with $n-1$ degrees of freedom.

Proof. Thm. 2.1

□

3 Examples

Example 3.1. For $T \sim t_5$, find $P(-2 \leq T \leq 2)$.

♢

Example 3.2. Let $X_1, \ldots, X_{10} \sim N(50, 4)$. Compute the distribution of $\frac{\sqrt{10}(\bar{X}-50)}{S}$.

♢

Example 3.3. Suppose that a random sample of $n = 26$ rainfall measurements, $X_1, \ldots, X_n$, is taken from a normal distribution with unknown mean μ and unknown variance σ^2. We are interested in how far the sample mean $\bar{X}$ will be from the true mean μ. We will use the quantity

$$U = \frac{\sqrt{n}(\bar{X} - \mu)}{\sqrt{S^2}}$$

here. Find the 90th percentile of $\bar{X}$ in terms of μ and S^2.

$\diamondsuit$

Example 3.4. Suppose that we observe $n = 20$ cheese chunks with lactic acid concentrations $X_1, \ldots, X_n$ coming from a normal distribution with unknown mean μ and variance σ^2. Find a value of c such that

$$\mathbb{P}\left(\bar{X} \leq \mu + c\sqrt{S^2}\right)$$

$\diamondsuit$

4 t Table

Table of the t Distribution

If X has a t distribution with m degrees of freedom, the table gives the value of x such that $\Pr(X \leq x) = p$.

m	$p = .55$	.60	.65	.70	.75	.80	.85	.90	.95	.975	.99	.995
1	.158	.325	.510	.727	1.000	1.376	1.963	3.078	6.314	12.706	31.821	63.657
2	.142	.289	.445	.617	.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925
3	.137	.277	.424	.584	.765	.978	1.250	1.638	2.353	3.182	4.541	5.841
4	.134	.271	.414	.569	.741	.941	1.190	1.533	2.132	2.776	3.747	4.604
5	.132	.267	.408	.559	.727	.920	1.156	1.476	2.015	2.571	3.365	4.032
6	.131	.265	.404	.553	.718	.906	1.134	1.440	1.943	2.447	3.143	3.707
7	.130	.263	.402	.549	.711	.896	1.119	1.415	1.895	2.365	2.998	3.499
8	.130	.262	.399	.546	.706	.889	1.108	1.397	1.860	2.306	2.896	3.355
9	.129	.261	.398	.543	.703	.883	1.100	1.383	1.833	2.262	2.821	3.250
10	.129	.260	.397	.542	.700	.879	1.093	1.372	1.812	2.228	2.764	3.169
11	.129	.260	.396	.540	.697	.876	1.088	1.363	1.796	2.201	2.718	3.106
12	.128	.259	.395	.539	.695	.873	1.083	1.356	1.782	2.179	2.681	3.055
13	.128	.259	.394	.538	.694	.870	1.079	1.350	1.771	2.160	2.650	3.012
14	.128	.258	.393	.537	.692	.868	1.076	1.345	1.761	2.145	2.624	2.977
15	.128	.258	.393	.536	.691	.866	1.074	1.341	1.753	2.131	2.602	2.947
16	.128	.258	.392	.535	.690	.865	1.071	1.337	1.746	2.120	2.583	2.921
17	.128	.257	.392	.534	.689	.863	1.069	1.333	1.740	2.110	2.567	2.898
18	.127	.257	.392	.534	.688	.862	1.067	1.330	1.734	2.101	2.552	2.878
19	.127	.257	.391	.533	.688	.861	1.066	1.328	1.729	2.093	2.539	2.861
20	.127	.257	.391	.533	.687	.860	1.064	1.325	1.725	2.086	2.528	2.845
21	.127	.257	.391	.532	.686	.859	1.063	1.323	1.721	2.080	2.518	2.831
22	.127	.256	.390	.532	.686	.858	1.061	1.321	1.717	2.074	2.508	2.819
23	.127	.256	.390	.532	.685	.858	1.060	1.319	1.714	2.069	2.500	2.807
24	.127	.256	.390	.531	.685	.857	1.059	1.318	1.711	2.064	2.492	2.797
25	.127	.256	.390	.531	.684	.856	1.058	1.316	1.708	2.060	2.485	2.787
26	.127	.256	.390	.531	.684	.856	1.058	1.315	1.706	2.056	2.479	2.779
27	.127	.256	.389	.531	.684	.855	1.057	1.314	1.703	2.052	2.473	2.771
28	.127	.256	.389	.530	.683	.855	1.056	1.313	1.701	2.048	2.467	2.763
29	.127	.256	.389	.530	.683	.854	1.055	1.311	1.699	2.045	2.462	2.756
30	.127	.256	.389	.530	.683	.854	1.055	1.310	1.697	2.042	2.457	2.750
40	.126	.255	.388	.529	.681	.851	1.050	1.303	1.684	2.021	2.423	2.704
60	.126	.254	.387	.527	.679	.848	1.046	1.296	1.671	2.000	2.390	2.660
120	.126	.254	.386	.526	.677	.845	1.041	1.289	1.658	1.980	2.358	2.617
∞	.126	.253	.385	.524	.674	.842	1.036	1.282	1.645	1.960	2.326	2.576

Table III, "Table of the t Distribution" from STATISTICAL TABLES FOR BIOLOGICAL, AGRICULTURAL, AND MEDICAL RESEARCH by R.A. Fisher and F. Yates. © 1963 by Pearson Education, Ltd.